

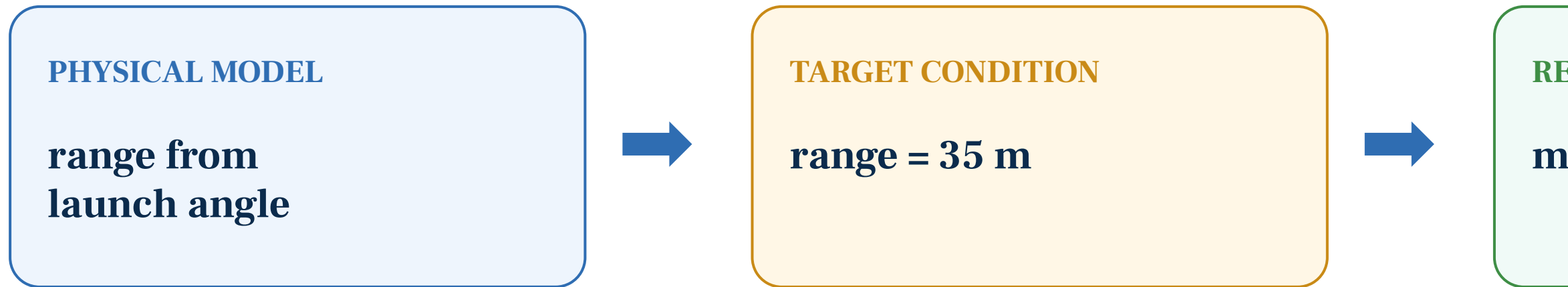
PHY4605 Computational Methods in Physics

From a Physical Target to a Checkable Root

Week 5 | Turning target conditions into numerical evidence

Turn the target condition into a residual

The unknown is an angle; the mismatch is a distance

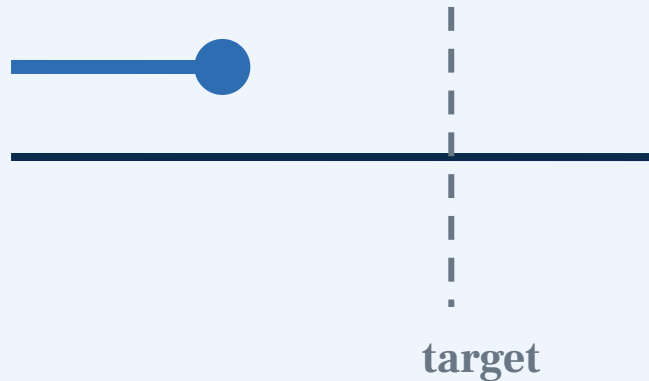


Read the residual sign as physics

A sign tells you which side of the target the model sits on

undershoot

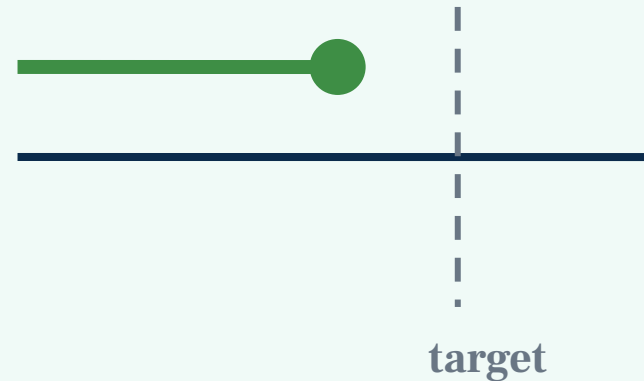
model range < target



negative

root

model range = target



zero

overshoot

model range > target

Find the zero crossing before coding

The graph suggests a bracket before the loop begins

low-angle bracket

Choose endpoints with
opposite residual signs

$$f(20 \text{ degree}) = -8.7905 \text{ m}$$

$$f(40 \text{ degree}) = +5.1553 \text{ m}$$

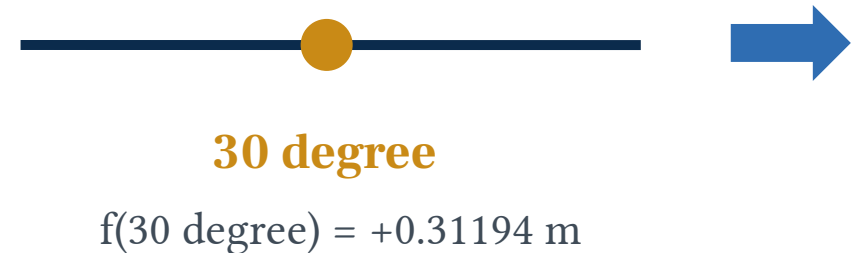
Bisection keeps a sign change alive

Midpoint $\rightarrow$ residual $\rightarrow$ retain the sign-changing half

current bracket



test the midpoint



1 choose opposite signs

2 compute midpoint

3 evaluate residual

Trace the first four bisection updates

Each row shrinks the interval because the sign change survives

iteration	current bracket (degree)	midpoint (degree)	f(mid) (m)	keep
1	[20, 40]	30	+0.31194	[20, 30]
2	[20, 30]	25	-3.7648	[25, 30]
3	[25, 30]	27.5	-1.5993	[27.5, 30]
4	[27.5, 30]	28.75	-0.61095	[28.75, 30]

the



Read the bisection scaffold, line by line

The code is the earlier reasoning written in MATLAB

1 MIDPOINT

centre the current bracket

2 RESIDUAL

measure range – target

3 STOP

use $\text{abs}(f_{\text{mid_m}}) < \text{tolerance_m}$

```
bis_left_deg = left_deg;
bis_right_deg = right_deg;
for iteration = 1:max_iterations
    mid_deg = (bis_left_deg + bis_right_deg)/2;
    f_mid_m = (v0_mps^2/g_mps2)*sind(2*mid_deg) - target_m;
    if abs(f_mid_m) < tolerance_m
        break
    end
    f_bis_left_m = (v0_mps^2/g_mps2)*sind(2*bis_left_deg) - target_m;
    if f_bis_left_m*f_mid_m < 0
        bis_right_deg = mid_deg;
    else
        bis_left_deg = mid_deg;
```

Stop once, then substitute back

A loop stop is a decision; substitution is the validation

bisection result

29.5679 degree

after 13 recorded iterations

one stopping rule

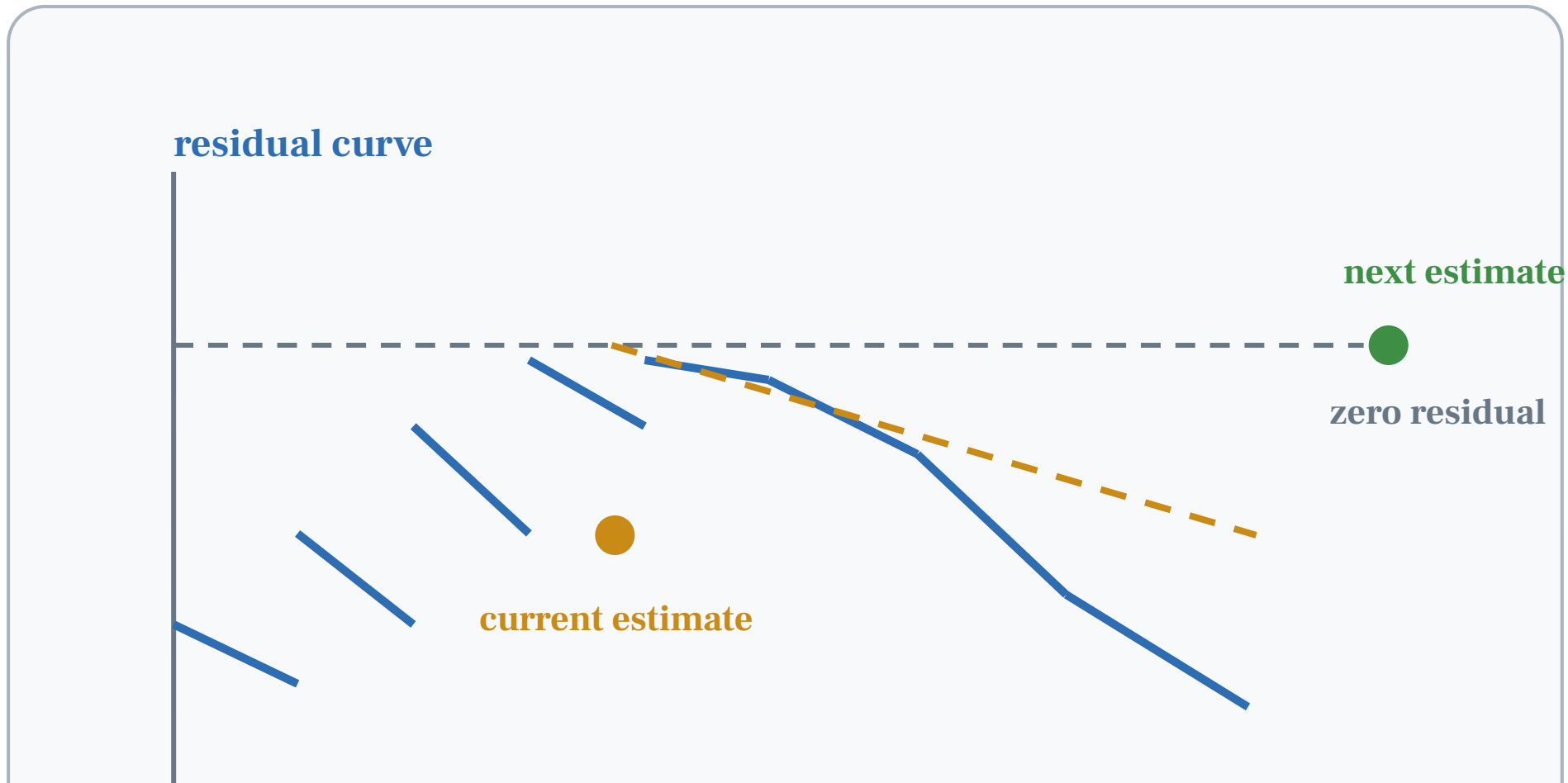
$\text{abs}(\text{residual}) \cdot 10^{-3} \text{ m}$

substitute into the original model

check	result
root angle	29.5679 degree

Newton follows the local tangent

A slope turns one current estimate into a next estimate



unit check

Trace Newton from 25 degree

The trace records exactly what each tangent update needs

```
newton_theta_deg = newton_start_deg;
newton_history = zeros(max_iterations,4);
for iteration = 1:max_iterations
    f_current_m = (v0_mps^2/g_mps2)*sind(2*newton_theta_deg) - target_m;
    slope_m_per_deg = (v0_mps^2/g_mps2)*cosd(2*newton_theta_deg)*(2*pi/180);
    newton_history(iteration,:) = [iteration, newton_theta_deg, f_current_m, slope_m_per_deg];
    if abs(f_current_m) < tolerance_m
        break
    end
    newton_theta_deg = newton_theta_deg - f_current_m/slope_m_per_deg;
end
newton_root_deg = newton_theta_deg;
```

iteration	theta (degree)
1	25
2	29.1
3	29.5
4	29.5

Working exposure: same root, different root

Different evidence structures can lead to the same physical answer

BISECTION

keeps a sign-changing bracket

13

recorded iterations

safety evidence

NEWTON

uses a local res

4

recorded

local efficiency

A midpoint is not enough: the bracket must change

Arithmetic can start a midpoint; opposite signs justify the method

SAME-SIGN BRACKET



midpoint exists



inspect → repair →
start

SIGN-CHANGE



bracket [20, 40]

Transfer the residual pattern to new physics

The equation changes; the reasoning chain stays stable

THIN LENS

root variable

image distance v (m)

residual unit

m^{-1}

define $\rightarrow$ bracket $\rightarrow$ update $\rightarrow$ check

Exit ticket — explain a root to a future you

Retrieve the reasoning chain without relying on function names

1

DEFINE

What target condition
does $f(x) = 0$ encode?

write one sentence

2

BRACKET

Why do endpoint
signs matter?

write one sentence

3

UPDATE

What does a midpoint
or tangent change?

write one sentence